

Ryan Nepal

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EDUCATION

University of Massachusetts, Amherst

Expected May 2027

B.S. in Mechanical Engineering

Coursework: Design of Mechanical Components, Fluid Mechanics, Manufacturing Processes, Thermodynamics, Dynamics, Fundamentals of Electrical Engineering, Strength of Materials, Materials Science, Statics, Ordinary Differential Equations

EXPERIENCE

Amazon | Reliability and Maintenance Engineering Intern

June 2026 - August 2026

Las Vegas, NV

- Designed and validated a PLC-controlled tote-turner using pneumatic actuators and photoelectric sensors to automate tote orientation on conveyance; leveraged industrial automation, ladder logic, and GD&T — increased accuracy 14%
- Built and deployed a full-stack web application on AWS (Elastic Beanstalk, Aurora PostgreSQL, S3) for tracking conveyor sorter amnesty across a site; first-of-its kind system enabling data-driven root cause analysis.
- Owned DMAIC project from baseline (1,171 units across 12 segments) through launch; 700+ units logged in first 10 days with projected 2-4 hours/week in time savings through elimination of ad hoc investigation and jam prevention.

NexTek, LLC | Mechanical Engineering Intern

May 2025 - August 2025

Billerica, MA

- Owned design, assembly, and validation of a leak test fixture to withstand 680 lbf for IP-rated products; performed FEA to evaluate assembly strength; tested 21,000+ units and increased efficiency by 400%.
- Supported end-to-end design and validation of EMI/RF protection components; applied CAD/3D modeling, GD&T, assembly, and bench testing with data acquisition to verify performance and reliability.
- Managed ~4 concurrent projects; led troubleshooting/failure analysis and drove ECOs/change control to resolve build issues and maintain spec compliance.

PROJECTS

Rebuild of a Salvage 2008 Mazda3 S

January 2026

- Performed a comprehensive mechanical and structural assessment, identifying front-end collision damage and engine performance issues through component inspection and fault-code diagnostics.
- Reconstructed the front-end assembly and engineered a repair for sheared intake manifold ports by precision tapping and installing vacuum valves and lines with high-temp epoxy; restored proper airflow and eliminated engine misfires.

Design, Simulation, and CNC Machining of a Fuel Pump Adapter

November 2025

- Designed a custom aluminum fuel pump adapter to mechanically interface an NPT hose connector with a pressurized fuel reservoir and fuel pump; used Ansys to validate structural integrity through pressure and stress analysis.
- Programmed and CNC-machined the adapter from stock material, performing deburring, fitment verification, and pressure testing to validate leak-free operation.

SKILLS

Software: SolidWorks | ANSYS | MATLAB | Inventor | Fusion | AutoCAD | Revit | PLC Programming | Microsoft Excel

Engineering: CAD / 3D Modeling | DFMA | FEA | CNC Machining | GD&T | Root Cause Analysis | DMAIC | Prototyping

LEADERSHIP

UMass Supermileage Vehicle Team | Fuel System Lead

September 2025 - Present

- Collaborate with a multidisciplinary engineering team to design, develop, and optimize a high-efficiency vehicle for competition at the Shell Eco-marathon at the Indianapolis Motor Speedway.
- Owned design and optimization of fuel delivery system; utilized CAD and FEA to improve combustion efficiency and structural integrity; applied CNC machining to develop efficient components; achieved 414 mpg in competition.

UMass Transit Services | CDL Instructor, Dispatcher, Operator

September 2024 - Present

- Train Class B CDL-permit drivers in safe operation, defensive driving, and transit procedures while instructing on bus mechanical systems to ensure thorough understanding of the vehicle.
- Direct day-to-day operations of a 28-bus fleet, supervising drivers, coordinating dispatch, and resolving service disruptions to ensure safe, reliable, and on-time transit service.